

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Final Term Examination 2025

Class: BSCS

Semester: V

Paper: Advance Database Systems

Max Marks: 50

Session (2023-2027)

Time Allowed: 3 Hours

Student Name: _____

Roll No: 095

Subjective Section (Marks = 35)

Q 1) Write short and precise answers.

(a) Write a MongoDB query to insert a new document into a students collection with fields: name, age, department, and GPA.

PLO	CLO	Bloom Taxonomy
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PLO-2	CLO-3	C3
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(b) Write a MongoDB query to delete all documents from the orders collection where the order status is "canceled".

PLO-2	CLO-3	C3
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(c) Write a MongoDB query to retrieve only the name and email fields for all customers in the customers collection.

PLO-2	CLO-3	C3
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(d) What is the difference between symmetric and asymmetric encryption?

PLO-2	CLO-2	C2
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Q 2) Write a PL/SQL block that accepts a department number and prints the total salary of all employees in that department. Use an aggregate function inside SELECT INTO. Handle NO_DATA_FOUND and OTHERS exceptions (10)

PLO-2	CLO-2	C3
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Q 3) Define ACID properties of a transaction in detail. Explain how each property ensures reliable and consistent database operations (5)

PLO-2	CLO-1	C2
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Q 4) Explain the importance of indexing in physical database design. Discuss different types of indexes (primary, secondary, clustered, non-clustered, composite, and bitmap) and how each affects query processing and performance tuning. (10)

PLO-2	CLO-2	C3
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